FRANCK PORTIER
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M1 MACRO
Group B
Session 10

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

When $y_t = \bar{y}$

then $c_t = \boxed{rA_t + \bar{y}}$

so that BC: $A_{t+1} = (1+r)A_t + \bar{y} - c_t$

$$A_{t+1} = (1+\cancel{r})A_t + \cancel{\bar{y}} - r\cancel{A_t} - \cancel{\bar{y}}$$

$$A_{t+1} = A_t$$

Undetermined coeff. method

- Guess a c. function
- Check that it satisfies optimality conditions

Assume $c_t = \phi_1 A_t + \sum_{i=0}^{\infty} \phi_{i2} y_{t+i}$

Assume $y_{t+i} = \bar{y} \ \forall i$

$\Rightarrow$ Guess $c_t = \phi_1 A_t + \phi_2 \bar{y}$

Optimality conditions

① Euler eq: $c_t = c_{t+1}$

$\Longrightarrow \cancel{\phi_1} A_t + \cancel{\phi_2 \overline{y}} = \cancel{\phi_1} A_{t+1} + \cancel{\phi_2 \overline{y}}$

$\Longrightarrow A_{t+1} = A_t$

② BC: $A_{t+1} = (1+r)A_t + \overline{y} - c_t$

$\Leftrightarrow \underbrace{\cancel{A_{t+1}}}_{=A_t} = (\cancel{1}+r)A_t + \overline{y} - [\phi_1 A_t + \phi_2 \overline{y}]$

$$(r - \underbrace{\phi_1) A_t}_{0} + \underbrace{(1-\phi_2)}_{0} \bar{y} = 0 \quad \forall A_t \quad \forall \bar{y}$$

$$\Rightarrow \begin{cases} \phi_1 = r \\ \phi_2 = 1 \end{cases}$$

$\Rightarrow$ With a guess $c_t = \phi_1 A_t + \phi_2 \bar{y}$,

I find $\boxed{c_t = r A_t + \bar{y}}$

Adaptive expectations (error-correcting mechanism)

$X^e_{t,t+1}$ = expected value of X_{t+1}, formed in period t

$$X^e_{t,t+1} = \alpha X^e_{t-1,t} + (1-\alpha)\underbrace{(X_t - X^e_{t-1,t})}_{\text{yesterday forecast error}}$$

$\alpha \in [0,1]$

...

Rat. Expectations

Define Expectation error:

$$\varepsilon_{t+1} = X_{t+1} - E_t(X_{t+1})$$

Property 1: $E_t[\varepsilon_{t+1}] = E_t\left[X_{t+1} - E_t[X_{t+1}]\right]$

$$= E_t[X_{t+1}] - \cancel{E_t}\left[E_t[X_{t+1}]\right]$$

$$= 0$$

$$AR(1): \quad X_t = \rho X_{t-1} + \varepsilon_t \qquad |\rho| < 1$$

$$\rho > 0$$

$$E_t(X_{t+1}) = E_t(\rho X_t + \varepsilon_{t+1})$$

$$= \rho \underbrace{E_t(X_t)}_{X_t} + \underbrace{E(\varepsilon_{t+1})}_{0}$$

$$= \rho X_t$$

$$\boxed{E_t(X_{t+j}) = \rho^j X_t}$$

$$E_t(X_{t+2}) = E_t[\rho X_{t+1} + \varepsilon_{t+2}] = E_t\left[\rho[\rho X_t + \varepsilon_{t+1}] + \varepsilon_{t+2}\right]$$

$$= E_t(\rho^2 X_t) + \rho E_t(\varepsilon_{t+1}) + E_t[\varepsilon_{t+2}]$$

$$= \rho^2 X_t$$